

MECHANICAL ENGINEERING

Instruction Book No. 4j1

For M. E. Numbers: 1667

MASTER COPY

LEPEL HIGH FREQUENCY INDUCTION
HEATING UNIT

Not For Check-out

MODEL NO. T-10-3-KC-F1

SERIAL NO. (RP)

TYPE NO. T-1001-237

MANUAL NO. 13190016

INSPECTION CERTIFICATE

The above unit, prior to shipment, was inspected and tested in accordance with the LEPEL Quality Control Standards and Procedures; and was found in all respects to be in proper operating condition.

LEPEL HIGH FREQUENCY LABORATORIES, INC.

Date _____

By: _____

Inspector

LepeL HIGH FREQUENCY LABORATORIES, Inc.

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T-10-3-KC-F1
T-10-3-KC-F2

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ADDENDUM

1. SAFETY

Since high frequency voltages can produce arcs and burns, operators must be instructed not to touch the load coil or load leads.

Appropriate guards should be provided to preclude the possibility of such contact. Since work-pieces may be very hot both during, and for some time after heating, the usual caution associated with such objects should be carefully observed.

NOTE: Where load coil lengths greater than two feet are required Lepel can supply coaxial lines which will meet both safety and electrical requirements.

SECTION 1

INTRODUCTION

1.1

GENERAL

Lepel induction heating generators are designed specifically for industrial heating applications. They conform to NEMA and IEEE standards for induction heating equipment and also follow JIC standards.

All Lepel generators are certified to meet the Federal Communications Commission Regulations, Part 18, subpart D, titled, "Industrial Heating Equipment".

1.2

DESCRIPTION

The basic elements of the generator are: control circuit, high voltage DC power supply, power control and power oscillator. The control circuit includes the switches monitoring lamps and relays necessary to operate and protect the equipment. The high voltage DC power supply includes the high voltage plate transformer and rectifiers used to convert the AC line voltage to a high DC voltage. The power control adjusts the RF output power by varying the output voltage of the DC power supply (See section 3.9 for more on the power control). The oscillator, which is a modified Hartley type, converts the DC power to RF power.

1.2 (Cont'd)

Taps on the tank coil and continuously variable grid drive give the flexibility to accommodate a wide range of loads and load coils.

1.2.1 WATER TO WATER HEAT EXCHANGER AND COOLING RECIRCULATOR (OPTIONAL)

The optional built in cooling water recirculator is designed for use in areas where the available local water supply is unsuitable for direct cooling of the equipment. Any of these conditions can make the water unsuitable: high mineral content, low resistivity, erratic pressure and very low water temperature.

The system consists of a reservoir tank for distilled water to be recirculated, a water to water heat exchanger to transfer heat from the recirculated water to the local cooling water, a pump to circulate the distilled water and a thermostically operated valve on the local water supply which reduces water consumption when the power dissipation is low. Controls include a protective interlock in case of low reservoir water level, a front panel warning light for low water and a front panel switch for the recirculator water pump. All parts in contact with the distilled water are non-ferrous and non-corrosive.

1.2.2 CLOSED CABINET (OPTIONAL)

The closed cabinet option is designed for situations where it is undesirable to use the air in the vicinity of the generator

1.2.2 (Cont'd)

for cabinet cooling, for example where the air is dust laden or mixed with corrosive fumes.

The system consists of gasketed cabinet panels and an internal air to water heat exchanger through which cabinet air is directed by fans.

1.3 SPECIFICATIONS

Enclosure:	Heavy gauge aluminum cabinet mounted on rigid steel base with industrial type casters.
Dimensions, F1:	34" wide, 72" high, 52" deep.
F2:	40" wide, 74" high, 60" deep.
Weight:	1600 lbs (Approx.)
Power Input:	Approximately 24 KVA, 90% power factor 230 volts, or 460 volts, or 550 volts (See note.)
Power Output:	10 Kilowatts into a magnetic steel calorimeter. Rated for continuous duty and tested in accordance with NEMA and IEEE standards.
Output Frequency:	200 to 450 KHz
Oscillator Tube:	One Type F-5606 triode.

1.3

(Cont'd)

Cooling Water Supply: 10 GPM, 40 PSI to 60 PSI
(without recirculator)

86°F, maximum inlet temperature

NOTE: Generators are manufactured in accordance with customer specifications to operate on any one of the listed voltage and frequencies. The nameplate attached to the generator will show the correct voltage and frequency for each unit.

1.3.1

SPECIFICATION - OPTIONAL

WATER TO WATER HEAT EXCHANGER AND COOLING RECIRCULATOR

Pump Capacity : 25 GPM at 45 PSI

Pump Motor (Ball
Bearing): 1-1/2 HP - 3600 RPM
3 Phase - 220/440 Volt
60 Cycle

Heat Exchanger Area: Square Feet

Reservoir Tank Capacity: 10 Gallons - Distilled Water

Cooling Water Supply: 12 GPM, 30 to 45 PSI, 80°F max. inlet temp.

1.3.2

SPECIFICATIONS - OPTIONAL SEALED CABINET WITH AIR
TO WATER HEAT EXCHANGER

Exchanger Area: 404 Square inches

SECTION 2

INSTALLATION

2.1' UNPACKING AND INSPECTION

Upon arrival, the crates and containers containing the generator and its components should be opened carefully and inspected. Any damage which may have resulted from handling and transportation must be reported at once to the carrier and Lepel High Frequency Laboratories, Inc.

Tubes are packed separately, using special materials for absorbing transportation shocks. Nevertheless, rough handling can cause internal defects, so tubes should be checked immediately upon delivery. An ohmmeter is adequate for making preliminary tests. Between filament terminals, the resistance should be very low (see oscillator tube terminal drawing).

Conversely, when measuring between filament and grid, filament and anode, or grid and anode, the resistance should be very high, typically in excess of several megohms. Again, any defects must be reported to the carrier and Lepel.

In addition to electron tubes, the other parts shown on the invoice should be examined and inventoried before starting the installation.

2.1 (Cont'd)

Remove the cabinet side panels and remove all cord and packing materials used to eliminate vibration and abrasion in transit. Do not remove the drive cord connecting the "GRID CONTROL" knob to the grid coil.

2.2 ELECTRICAL

2.2.1 POWER LINE CONNECTION

The generator should be connected to a three-phase 60 Hertz power line through a fusible disconnect switch. Requirements are:

Line Voltage - Nominal	230V	460 V	550V
Line Current - Full Load	70A	35 A	25 A
Phases	3	3	3
Cable Size	2	6	8
Fuses	110 A	60 A	40A

From the disconnect box, run the power cables through the access hole in the bottom of the rear cabinet panel. Connect the power line conductors to the terminal board provided inside the cabinet and adjacent to the access hole.

2.2.2

GROUND CONNECTION

Ground posts are provided on the base frame of the cabinet. A No. 4 Wire or, preferably, a copper strip about 1/32" x 2" should be attached to one of these posts and connected to the nearest ground. It is important to make this connection as short as possible. This "ground", if it is a waterpipe should actually run into the ground at a point nearby. Overhead piping should be avoided. Where possible, a 8-foot copper or brass pipe may be buried vertically into the ground near the generator to serve this purpose.

2.3

COOLING WATER

2.3.1

WATER SUPPLY

The cooling water supply should deliver at least 10 gallons per minute at a pressure between 40 PSI and 60 PSI. (12 GPM and 30 PSI to 45 PSI if equipped with recirculator) It is best to have a generous supply since fluctuations in the flow could result in interruption of generator operation.

A water strainer is supplied with the equipment and should be installed at the generator water inlet. The system should also contain an inlet shut-off valve, a pressure gauge, and an auxiliary valve to supply water for fixtures and other accessories requiring a separate water circuit.

2.3.2 WATER DRAIN

The water outlets should lead to an open drain (gravity drain) near the generator to enable the operator to keep a visual check on the water flow through the circuit. Overhead returns should be avoided. The drain capacity must be greater than the actual consumption for obvious safety reasons.

If it should become necessary to resort to overhead drains, a check valve must be installed in the drain pipe near the generator. Never use a shut-off valve in the drain as failure to open this valve would defeat the purpose of any protective water pressure switch that may be incorporated in the system.

2.4 COMPONENT INSTALLATION

2.4.1 TUBE INSTALLATION

Carefully unpack the triode oscillator tube (Type F-5606). With the side panels off, the insulated bracket for supporting the tube can be seen in the top and towards the rear of the cabinet.

When installing the oscillator tube be sure the rubber gasket is properly seated and that the three screws holding the oscillator tube in the water jacket are firmly tightened. Connect the flexible filament leads from the coaxial feed line in the left rear corner to the filament posts of the oscillator tube. See Dwg. No. 6154-23.

2.4.1 (Cont'd)

The six sockets for rectifier tubes are on top of the unitized rectifier filament transformer box located in the bottom of the generator. Install the three thyatron tubes (type 678) in the sockets nearest the terminal strips for wiring connections. This is usually the side closest to the outside of the generator. Install the three rectifier diode tubes (type 575 -A) in the other three sockets. Connect anode cap wires per drawing No. 3730005, "Connection Diagram, Unitized Rectifier Bank.

Generators using primary saturable reactor, primary SCR, or control diode power control use either six rectifier diode tubes or six silicon diode assemblies. See the schematic drawing. If diode tubes are used they plug into the same filament transformer box used with the thyatrons. If silicon diodes are used they plug into fuse clips in the lower compartment of the generator, or in some cases, are pre-mounted in a six diode assemblies.

2.4.2 PLATE CHOKE AND RESISTOR INSTALLATION

The plate RF choke is the large coil wound on a five inch diameter form. It mounts horizontally and hangs down from the main shelf on porcelain insulators that can be reached from the left side. The two grid resistors R4 and R5 should now be mounted in the fuse clips on the bracket above the oscillator tube.

2.4.3 TANK CAPACITOR INSTALLATION

Mount the tank capacitor in the upper compartment. Water and electrical connections are made as shown in the tank capacitor installation diagram.

LEPEL FIELD SERVICE

Lepel High Frequency Laboratories, Inc. has a staff of skilled service engineers who are factory-trained to service induction heating equipment, to instruct operating personnel, and to aid in application engineering. Their services are also available to supervise installation. Where the services of a Lepel service engineer have not been provided for in the contract and they should be required, contact your Lepel representative before attempting to install the equipment. If the assembly of the equipment is supervised by the Lepel service engineer, he assumes full responsibility for setting up the initial operation of the equipment.

If his services were provided for in the contract, arrange with your Lepel representative for a mutually satisfactory date for the installation. Water and power lines must be connected, and fixtures, load coils and work samples should be available when he arrives.

SECTION 3 OPERATION

3.1 SAFETY

It is important that service and operating personnel be aware of the high voltages present within the enclosure of this generator, and that the following safety precautions be observed.

When in operation the generator panels must be in place and securely fastened.

Operators must be instructed not to touch the work coil or object being heated. High frequency coil currents can produce arcs and burns.

The generator panels have interlock switches which automatically turn off the very high voltage circuits. The hazards normally accompanying the service of AC power circuits, (460 volts or 230 volts for example), are still present when these switches are open, so only qualified personnel should be permitted to remove the generator panels and work on the circuits. It is recommended that the AC power service be disconnected from the equipment when general service or maintenance is to be performed.

3.1 (cont.)

WARNING!

High Voltages Are Present In Generator Cabinet

If, for any reason, safety interlock switches are bypassed or the generator is operated with a panel open, alternating and direct potentials in excess of 10,000 volts will be exposed. These voltages may be fatal upon contact.

Residual charges may remain in the capacitors for some time after the generator is shut off. These capacitors should be discharged before touching anything within the generator.

3.2 TRANSFORMER TAP SETTING

Measure the phase to phase line voltage at the power line disconnect switch. With this switch in the "OFF" position, move the taps on the plate, oscillator filament, and rectifier transformers (T1, T2 & T3) to the terminals which most nearly correspond to the measured line voltage. T1 has three movable links for this adjustment, T2 has one movable wire and T3 has six movable wires and each must be set at identical tap values. See drawing No. 3730005 for T3 connections. Generators equipped with silicon diode rectifiers will not have transformer T3. These generators have the code letter (J) or (T) after the serial number on the name plate. Also, generators with constant voltage filament transformers are not provided with adjustable taps.

3.3

WATER CHECK

Connect a load coil to the RF output terminals. Do not use pipe joint compound or teflon tape on the coil connections. Turn the solenoid switch on the front panel to the "OPEN" position. In the "CLOSED" position water does not flow through the load coil and plate power can not be switch on. Turn the external cooling water on gradually and check all water connections inside and outside of the generator for leaks. Make sure the drain water is flowing freely. If equipped with the optional built in recirculator the flow may be small until power is on for a while.

3.3.1 WATER RECIRCULATOR START-UP

If the generator is equipped with the optional built in recirculator fill the reservoir tank with distilled water to within two inches of the top.

The pump for the recirculating water has a three phase motor which is started by the "RECIRCULATOR PUMP" switch on the front panel. At the initial start-up it is very important that the direction of rotation of the motor shaft be checked visually to make sure that it coincides with the arrow on the outside of the pump housing. Turn on the main power at the external switch and turn on the pump to check rotation. If equipped with a pressure gauge the water pressure should be over 40 PSI if rotation is correct. If rotation is not correct turn off the pump and the external switch and interchange the connections of any two of the three phase power feed cables to the generator. Do not interchange power wires at the motor or motor starter because the thyatron power control or primary SCR power control (if so equipped) would still have the wrong phase rotation.

3.3.1 (Cont'd)

In other words if the feed cables are connected to give the correct rotation to the motor, the phasing will be correct for the power control.

After the pump is running properly check for water leaks in the generator and recheck the reservoir level.

3.4 FILAMENT VOLTAGE CHECK

With the cooling water flowing, switch on the main circuit breaker located on the front panel.

The red "WATER FLOW ADEQUATE" lamp should be off. Push the "FILAMENT-ON" pushbutton. The oscillator filament volt meter on the control panel should now read 22.5 volts plus or minus one volt. With the plate power "OFF" the rectifier tube filament voltage may be measured at the tube sockets with an external test meter. 12,000 volts DC is on these terminals when plate power is on! Make sure an interlock plug is out so that plate power will not be turned on accidentally. Rectifier filament voltage should be between 4.9 volts and 5.2 volts. If oscillator tube or rectifier tube voltage are not in the correct range adjust the voltage with the tap connections on the filament transformer T2 and T3. Move to a higher tap value if filament voltage is high and conversely to a lower tap value if voltage is low.

If the generator is equipped with optional constant voltage filament transformers, the filament voltages should be in the proper range and adjustments are unnecessary. Adjustable tap connections are therefore not provided. If equipped with silicon diode rectifiers there will not, of course, be a rectifier filament transformer.

3.5 MODE OF OPERATION

This generator can be controlled in a number of ways. These include remote pushbutton control, automatic temperature control, remote power control, pulsing power control, and main panel control. Selection of these modes is made with toggle switches mounted on the internal control wiring panel in the bottom and toward the front of the generator. Terminal board TB1 which is used for the connection of external devices is also on this panel.

If a remote pushbutton control is supplied set the "CONTROL SYSTEM" switch on "REMOTE". Connect the control cable to the generator terminal board TB1 using the terminals specified on the remote control schematic diagram.

When the "TEMP. CONTROL"* switch is in the "REMOTE" position generator power level is regulated by a zero to five milliamperes DC signal applied to the + and - terminals on TB1. Observe the polarity. This facilitates the use of external devices such as temperature sensors, set point units, and controllers for automatic power control.

With the "POWER CONTROL"* switch in the "REMOTE" position an external potentiometer can be used to adjust power level. This potentiometer is also connected to terminal board TB1. This is generally a 1000 Ohm potentiometer rated for 2 watts or more. See the generator schematic diagram for correct terminal numbers.

3.5

(cont'd)

When the "RF POWER" switch is in the "FOOTSWITCH" position power is turned on or off by closing or opening a pair of external switch contacts (such as a footswitch). The external switch is connected via the three prong twistlock receptacle on the generator; the center pin is used for grounding. On generators with T-660 thyatron power control, or with SCR primary power control, this mode allows the large plate contactor to be on (closed) continuously while power is switched on and off. This permits accurate timing and minimum wear and tear on short time cycles.

When all switches are in the "OFF" or "PANEL" position controls on the front panel are used to operate the generator.

*NOTE: This feature is not present on all Lepel generator models. See the generator schematic drawing to check. Generally available if the generator has one of the following types of power control. T-660 Thyatron, SCR Driver with primary saturable reactor, SCR driver with control diode, or SCR primary power control.

3.6

START UP PROCEDURE

After installing the mercury vapor rectifier tubes it is necessary to operate them with only filament voltage on for 30 to 60 minutes. This allows the liquid mercury which has splashed during transit to distribute itself properly in the tube envelope. After this, and completing all preceeding steps the generator is ready for a trial run.